Lakshya JEE 2.0 (2024)

Application of Derivative

DPP-05

1. Find the critical point of the function $f(x) = x \ln x$.

2. Number of critical points of the function $f(x) = \dfrac{x^2 - 2}{x}$

(1) 0
(2) 1
(3) 2
(4) None of these

3. The number of critical points for the function $f(x) = 10xe^{3-x^2}$ is

(1) 0
(2) 1
(3) 2
(4) 3

4. The number of critical points for the function $y = f(x) = 6x - 4\cos(3x)$ in $x \in [0, \pi]$ is

(1) 1
(2) 2
(3) 3
(4) 4

5. On $[1, e]$ the greatest value of $x^2 \log x$

(1) e^2
(2) $\dfrac{1}{e}\log\dfrac{1}{\sqrt{e}}$
(3) $e^2 \log\sqrt{e}$
(4) None of these

6. If m and M respectively the minimum and maximum of $f(x) = (x-1)^2 + 3$ for $x \in [-3, 1]$, then the ordered pair (m, M) is equal to:

(1) (–3, 19)
(2) (3, 19)
(3) (–19, 3)
(4) (–19, –3)

7. On the interval [0, 1] the function $x^{25}(1 - x)^{75}$ takes its maximum value at the point

(1) 0 (2) 1/4
(3) 1/2 (4) 1/3

8. The difference between the greatest and least values of the function $f(x) = \sin 2x - x$ on $[-\pi/2, \pi/2]$ is:

(1) π (2) $\dfrac{\sqrt{3}+\sqrt{2}}{2} + \dfrac{\pi}{6}$
(3) $\dfrac{\sqrt{3}}{2} + \dfrac{\pi}{3}$ (4) $\dfrac{\sqrt{3}+\sqrt{2}}{2} - \dfrac{\pi}{3}$

9. Let $f(x) = \cos \pi x + 10x + 3x^2 + x^3, x \in [-2, 3]$. The absolute minimum value of $f(x)$, is:

(1) 0 (2) –15
(3) $3 - 2\pi$ (4) None of these

10. The function $y = x^x$ has:

(1) No local minimum value
(2) A local minimum value at $x = \dfrac{1}{e}$
(3) No local maximum value
(4) A local maximum value at $x = \dfrac{1}{e}$

11. If $f(x) = x^5 - 5x^4 + 5x^3 - 10$ has local maximum and minimum at $x = p$ and $x = q$, respectively, then $(p, q) \equiv$

(1) (0, 1) (2) (1, 3)
(3) (1, 0) (4) None of these

12. The critical points of the function $f(x) = (x - 2)^{2/3}(2x + 1)$, are

(1) 1 and 2
(2) 1 and –1/2
(3) –1 and 2
(4) 1

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. $x = \frac{1}{e}$
2. (2)
3. (3)
4. (2)
5. (1)
6. (2)
7. (2)
8. (1)
9. (2)
10. (2, 3)
11. (2)
12. (1)